

CITS5508 Practice Exam — Chapter 4

Training Models

- This practice paper has **5 questions** for **60 marks**.
 - Allow approximately **two minutes per mark** (≈ 2 hours total).
 - Only UWA-approved calculators permitted.
 - Answers are in the separate file *Practice Exam - Chapter 4 (Answer Key).md*.
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Question 1. (12 marks)

- (a) A linear regression model can be trained using the **Normal Equation** or using **gradient descent**. State **two** situations in which you would prefer gradient descent over the Normal Equation, and explain why. (4 marks)
- (b) Describe the roles of the three gradient descent variants — **batch**, **stochastic**, and **mini-batch** GD — and state one advantage of stochastic GD over batch GD. (4 marks)
- (c) Explain the effect of the **learning rate** η on gradient descent if it is set (i) too small and (ii) too large. Why is feature scaling important before running gradient descent? (4 marks)
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Question 2. (12 marks)

- (a) Explain what a **learning curve** plots. Sketch or describe the shape of the training and validation curves for (i) a model that is **underfitting** and (ii) a model that is **overfitting**. (6 marks)
- (b) The generalisation error can be decomposed into **bias**, **variance**, and **irreducible error**. Define each, and state which one is reduced by increasing model complexity. (4 marks)
- (c) You train a ridge regression model and observe that the **training error and validation error are almost equal and both fairly high**. Is the model suffering from high bias or high variance? Should you **increase** or **decrease** the regularisation hyperparameter α ? Justify. (2 marks)
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Question 3. (12 marks)

- (a) Write down (in words or formula) how the cost function of **ridge regression** differs from that of **lasso regression**. (3 marks)
- (b) Lasso tends to produce a **sparse model** while ridge does not. Explain *why* lasso drives some weights exactly to zero whereas ridge only shrinks them. (4 marks)
- (c) For each situation, recommend ridge, lasso, elastic net, or plain linear regression, and justify: (3 marks) 1. You want at least a little regularisation and have no reason to suspect useless features. 2. You suspect only a few of many features are useful. 3. You have more features than instances and several features are strongly correlated.
- (d) Explain **early stopping** as a regularisation technique for iterative learners. (2 marks)
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Question 4. (12 marks)

A logistic regression model estimates the probability that a student **passes** a unit:

$\hat{p} = \sigma(\theta_0 + \theta_1 \cdot x_1 + \theta_2 \cdot x_2)$, where $\sigma(t) = 1 / (1 + e^{-t})$,

with x_1 = hours studied, x_2 = prior GPA, and fitted coefficients $\theta_0 = -4$, $\theta_1 = 0.05$, $\theta_2 = 0.8$.

(a) A student has a GPA of $x_2 = 3.0$. How many **hours** must they study to have a **50%** chance of passing? Show your working. (4 marks)

(b) For a student with GPA 3.0 who studies **40 hours**, compute the estimated probability of passing. (3 marks)

(c) Explain why setting the estimated probability to 50% corresponds to setting the **logit** (the linear part) to zero, and state what kind of **decision boundary** logistic regression produces. (3 marks)

(d) You must classify photos along two independent axes: indoor/outdoor **and** day/night. Should you use one **softmax** regression classifier or two **logistic** regression classifiers? Justify. (2 marks)

Question 5. (12 marks)

A dataset has been split and scaled as follows:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

(a) Implement a **ridge regression** model with scikit-learn. **Grid-search** the regularisation strength α over **five** values using **k-fold cross-validation with $k = 3$** , refit the best model on the whole training set, **print the best α** , and compute the **mean squared error on the test set**. (6 marks)

(b) In scikit-learn's LogisticRegression, the regularisation strength is controlled by C, not alpha. State the relationship between C and regularisation strength. (2 marks)

(c) Write down **all** the polynomial features (up to and including degree 3) that PolynomialFeatures(degree=3) would generate for two input features x_1 and x_2 (excluding the bias term). (4 marks)